Winter Weather Panel: Descriptive Report

PRISM county–month data, 1981–2025

Version 2

2026-10-22

Provenance. All exhibits generated by `codePYTHON/09_descriptive_weather_full.ipynb`, executed on Kodama against the complete PRISM county-month panel. Figures are vector PDFs reproduced without rescaling of content. Tier 1 QA exhibits are indexed in Appendix B rather than reproduced.

What changed since the 2026-08-28 report. Titles and source notes have been removed from the images and the notes now appear beneath each figure; map legends are horizontal and sit at lower left; state borders are drawn in black. New exhibits: all state trends pooled into one figure, a Data-section composite, a full-sample mean winter temperature map, an early/recent/change treatment of extreme-cold days, the Winter Severity Index, area-weighted companions to the national exhibits, and the cross-season anomaly and trend comparisons.

One definition changed. The warm-anomaly exhibits previously flagged a *detrended* residual above 1.5 SD. They now use the same construction as `warm_winter_1sd` and `warm_winter_2sd` in the merged county-year panel: the raw winter mean above the county's own full-sample mean by one or two standard deviations. The figures and the regression table therefore describe the same object.

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1 Coverage and data integrity

The panel is complete: no duplicate keys, no unmatched rows across the monthly and derived files, no incomplete county-months, and no QA-field mismatches. The excluded county-winters are exactly the two boundary winters that cannot be complete, not unexplained missingness.

Table 1: Coverage and integrity diagnostics

Metric	Value
dataset	PRISM
monthly rows	1678320
derived rows	1678320
merged rows	1678320
counties	3108
first year	1981
last year	2025
duplicate monthly keys	0
duplicate derived keys	0
monthly only rows	0
derived only rows	0
incomplete months	0
qa mismatch n days	0
qa mismatch ppt total	0
qa mismatch dataset types	0
qa mismatch expected days	0
qa mismatch is incomplete	0
candidate county winters	142968
complete county winters	136752
excluded incomplete county winters	6216
first complete winter	1982
last complete winter	2025
expected calendar years	1981–2025
observed calendar years	1981–2025
observed complete winters	1982–2025

Table 2: Tier 1 QA findings

Check	Result	Status
Duplicate county-year-month keys	0	PASS
Rows appearing in only one merge source	0	PASS
Incomplete county-months	0	PASS
Excluded incomplete county-winters	6216	INFO
County-variable mean flags ($\|z\| \geq 5$)	20	REVIEW

2 Long-run winter warming

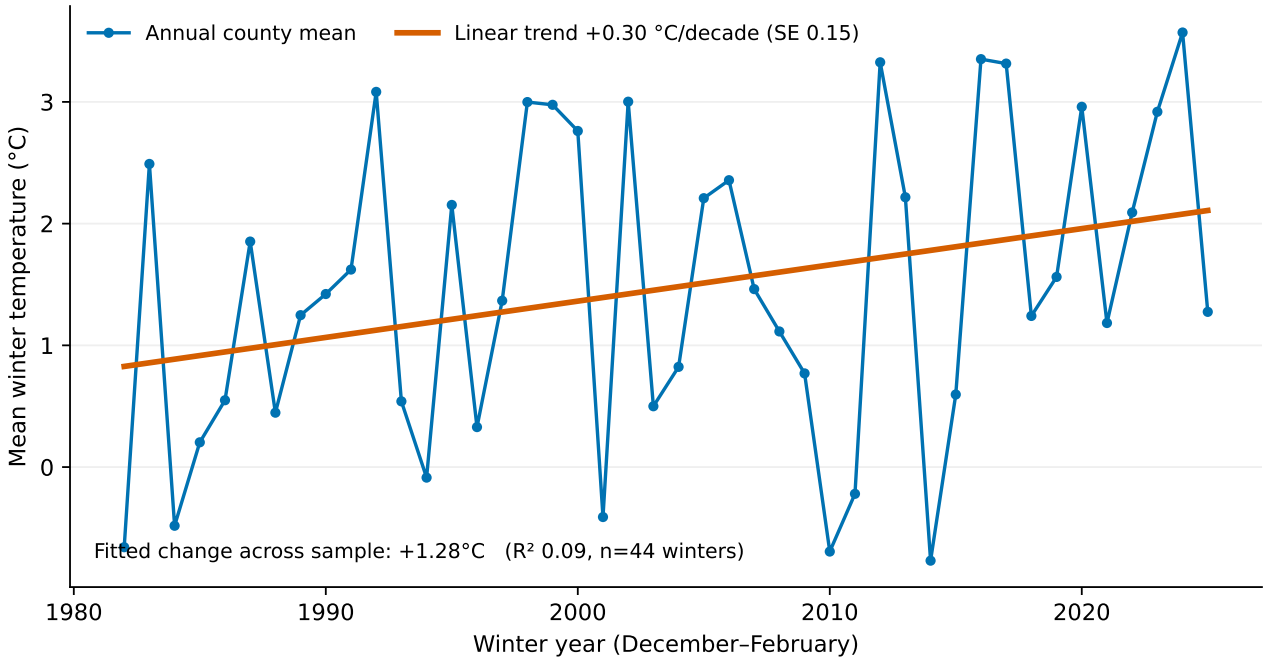


Figure 1: National mean winter temperature, 1982–2025, with a fitted linear trend.

Notes. Source: PRISM; complete DJF winters only. Estimand: the average county (counties weighted equally), not an area-, population- or road-weighted national average.

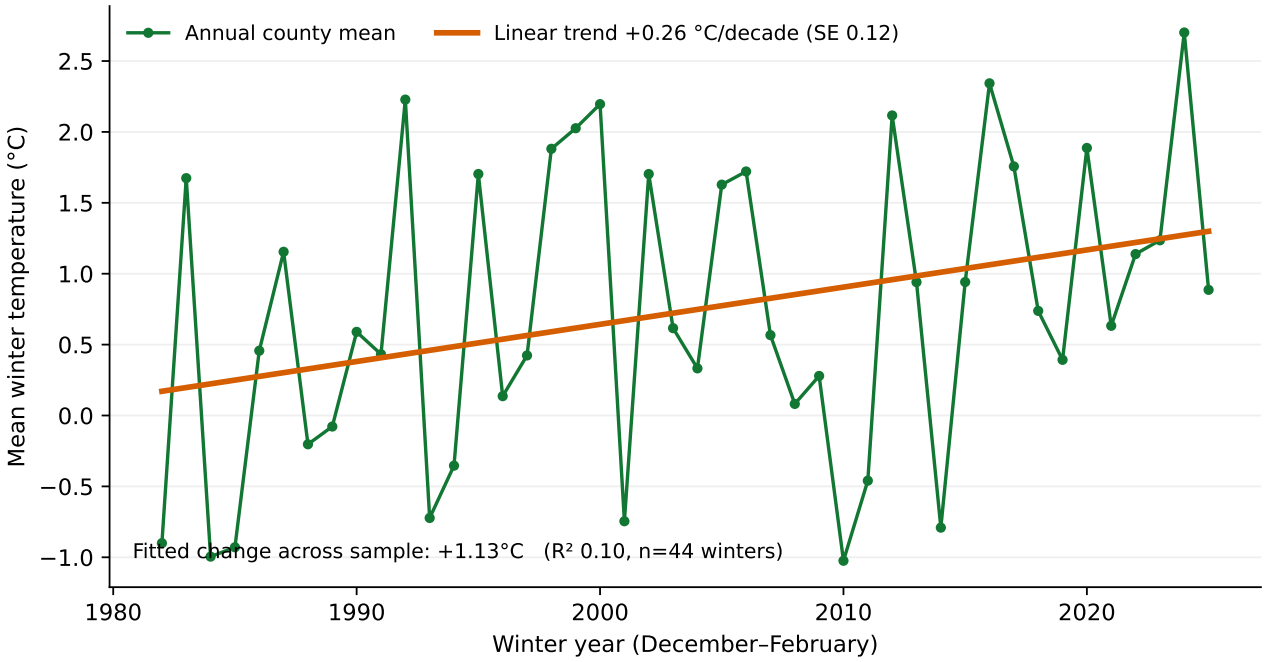


Figure 2: The same series with counties weighted by land area. The two answer different questions and both are reported.

Notes. Source: PRISM; complete DJF winters only. Estimand: the average square kilometre of the contiguous United States (counties weighted by land area, population weighting ruled out 8/28); counties in the west are much larger than counties in the east, so the unweighted and area-weighted series answer different questions.

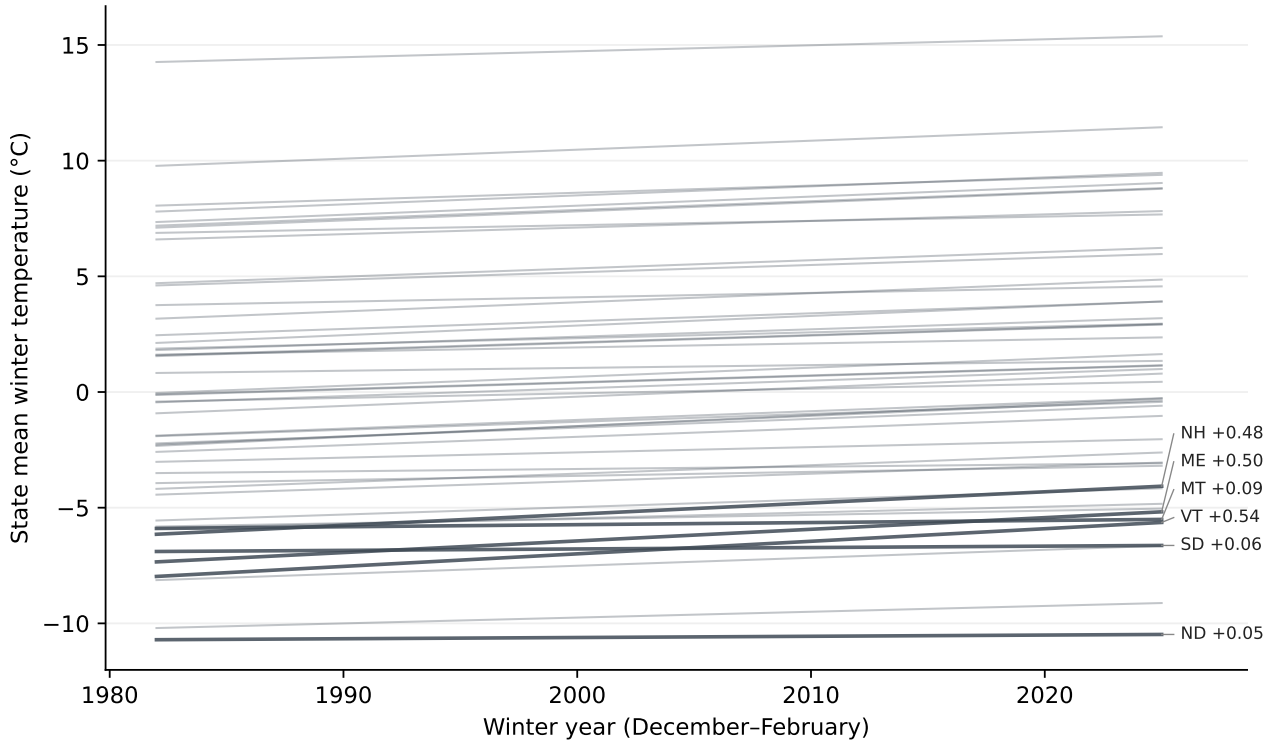


Figure 3: Every state's fitted winter-temperature trend in one figure. States differ in both level and slope.

Notes. Source: PRISM; complete DJF winters. One fitted linear trend per state (48 contiguous states plus DC), drawn in a single colour with transparency — the figure shows the spread of intercepts and slopes, and is not meant to make any individual state legible. Labelled: the three steepest warming trends (VT (+0.54), ME (+0.50), NH (+0.48)) and the three shallowest (ND (+0.05), SD (+0.06), MT (+0.09)), in °C per decade.

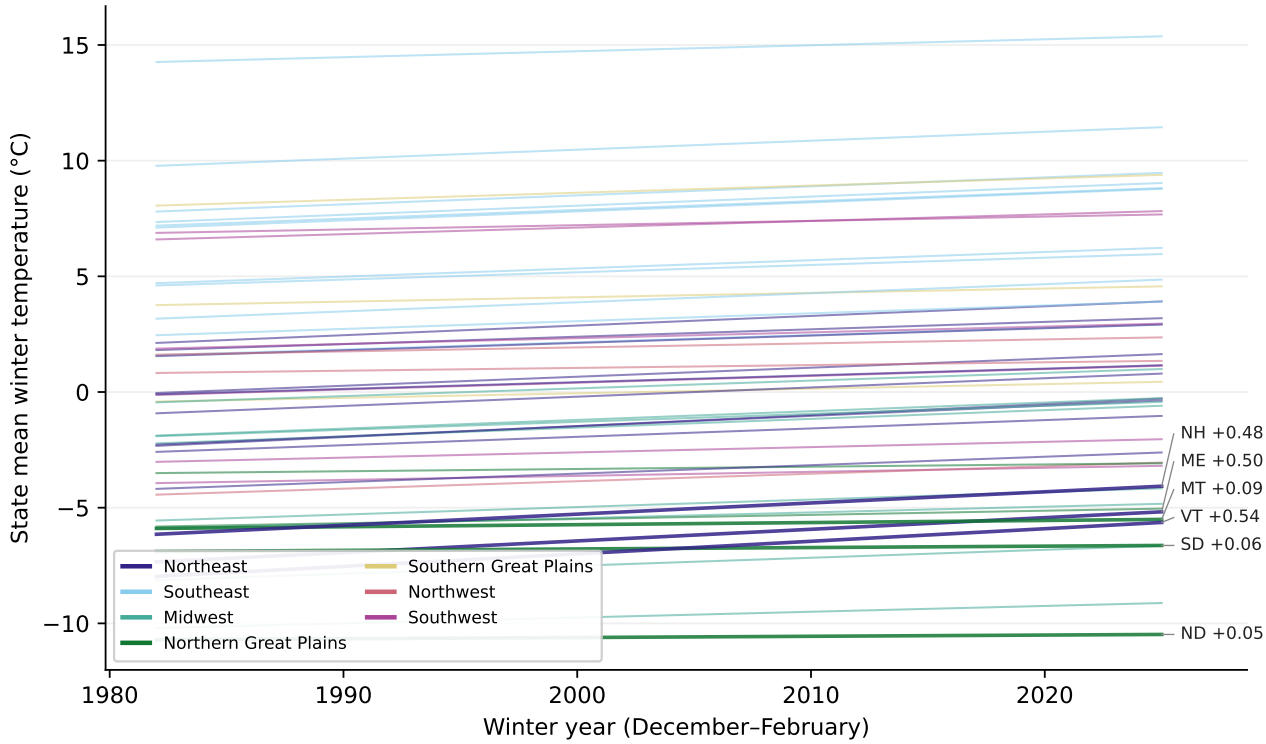


Figure 4: The same trends coloured by EPA climate-impact region.

Notes. Source: PRISM; the same fitted state trends coloured by EPA climate-impact region. Regions are assigned whole states, so no state is split; EPA Level I ecoregions cut across state lines and would need a county-level assignment.

State-level trends are reproduced panel by panel in Appendix A.

3 Where winters have warmed

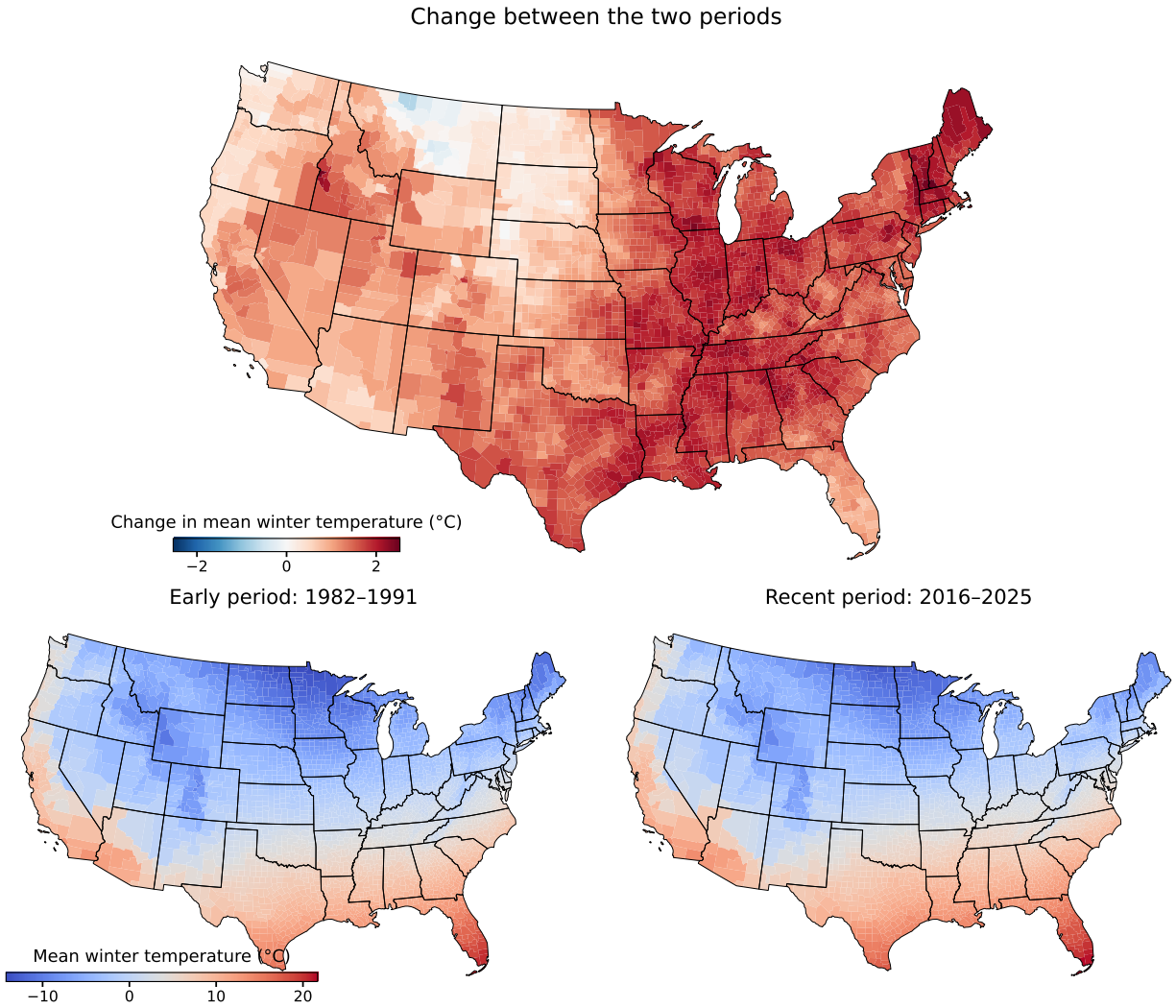


Figure 5: Proposed Data-section figure: the change between the two periods above, the two period means below.

Notes. Source: PRISM; complete DJF winters. Top: the change in mean winter temperature between the two periods, on a diverging scale centred at zero. Bottom: the two period means on a shared scale. Each county value is a mean over the 10 winters in its period (1982–1991 and 2016–2025), not a comparison of single winters or of sample endpoints. County borders are suppressed; state borders are drawn in black.

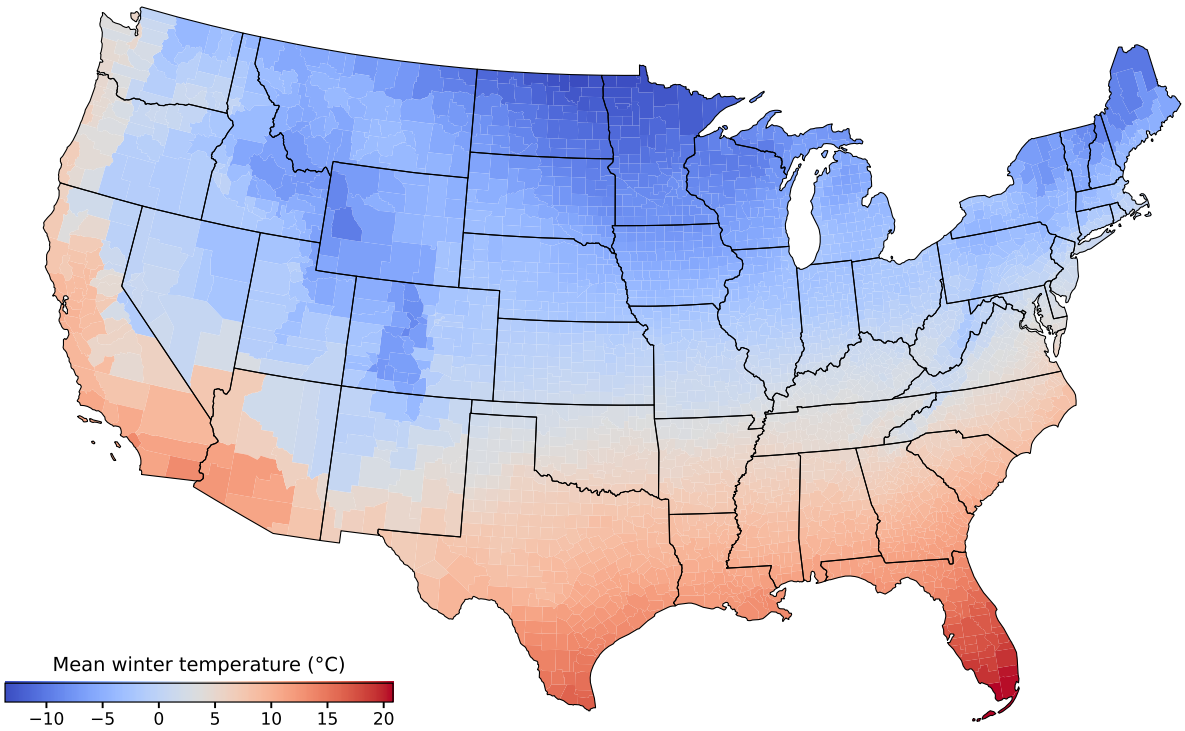


Figure 6: Mean winter temperature over the full sample – the climatology the anomaly measures are taken against.

Notes. Source: PRISM; mean of each county's complete DJF winters, 1982–2025. This is the climatology each county's warm-winter anomalies are measured against, so it is the baseline the anomaly maps standardise away.

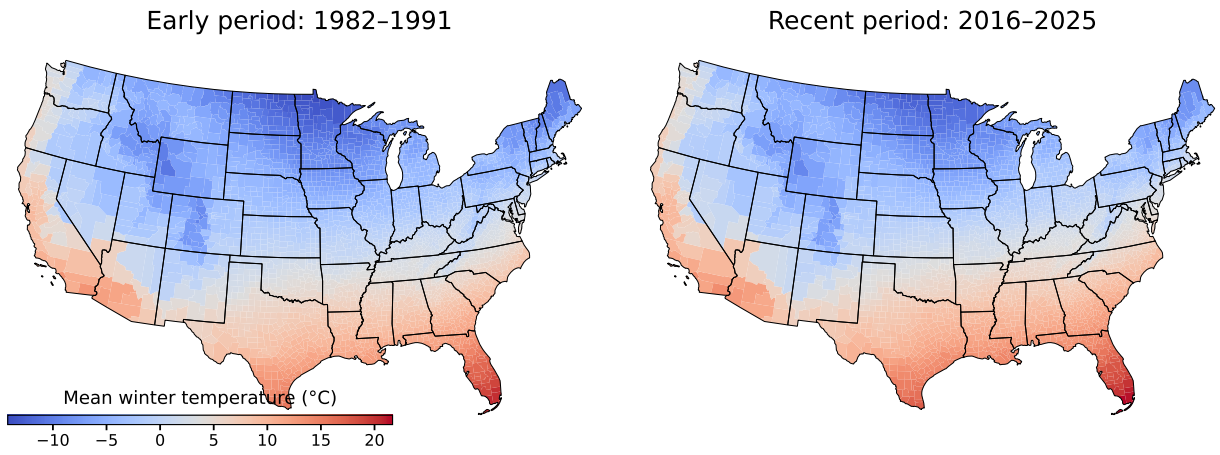


Figure 7: Mean winter temperature in the early and recent decades, on a shared colour scale.

Notes. Source: PRISM; complete DJF winters. Both panels share one colour scale, so they are directly comparable. County borders are suppressed; state borders are drawn in black. Each county value is a mean over the 10 winters in its period (1982–1991 and 2016–2025), not a comparison of single winters or of sample endpoints.

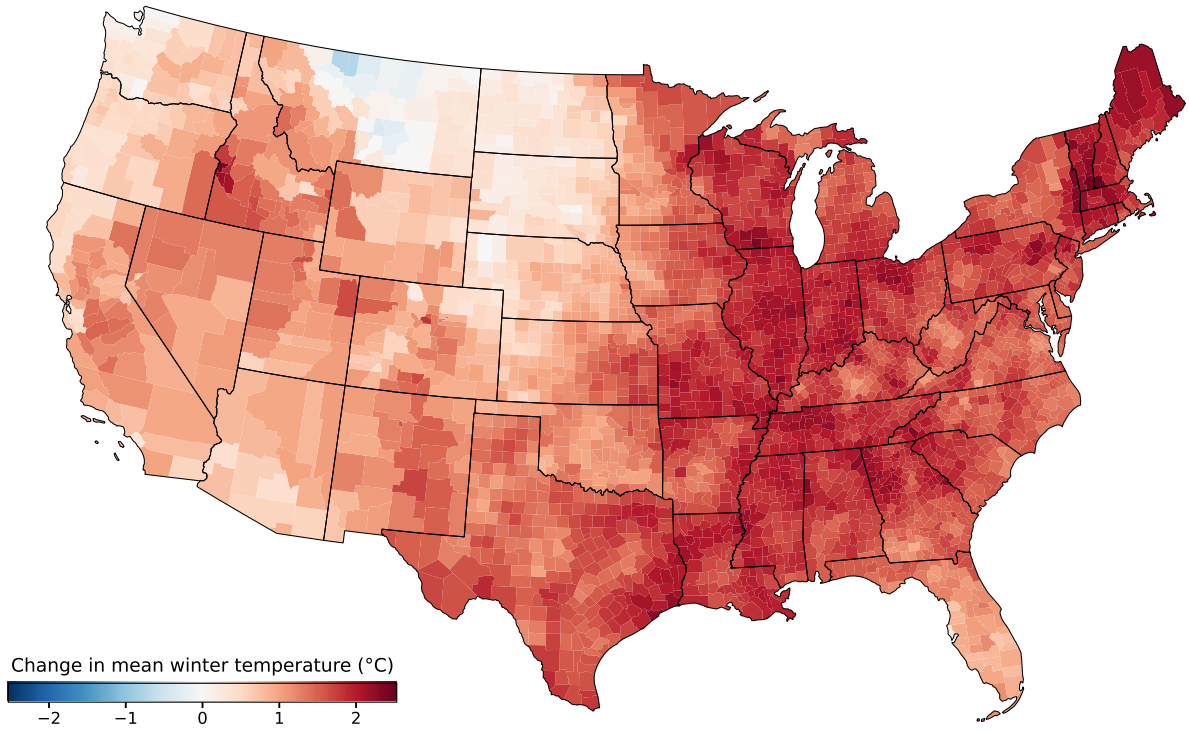


Figure 8: Change in mean winter temperature between the two periods.

Notes. Source: PRISM; complete DJF winters. Red is warming, blue is cooling, on a diverging scale centred at zero. The comparison is 2016–2025 minus 1982–1991. Each county value is a mean over the 10 winters in its period (1982–1991 and 2016–2025), not a comparison of single winters or of sample endpoints.

4 Winter cold extremes

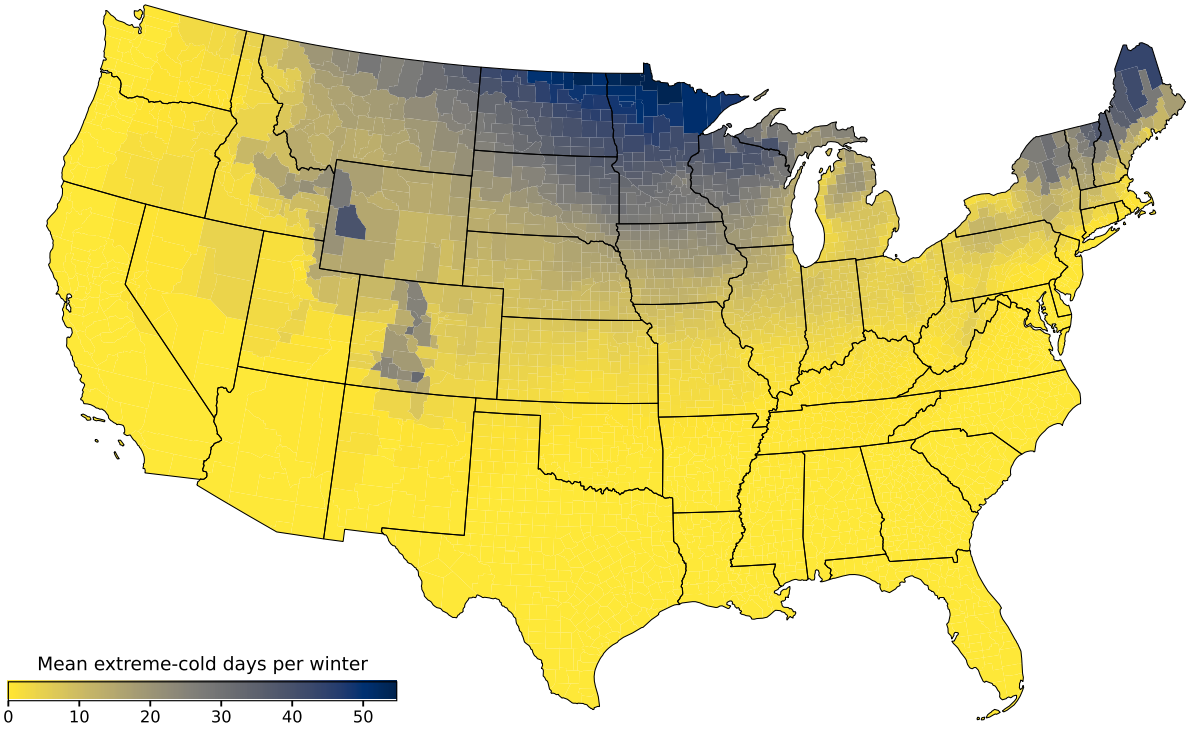


Figure 9: Mean count of days from 1 December to 30 April with a daily minimum at or below 0°F.

Notes. Source: PRISM; days from 1 December to 30 April with a daily minimum at or below 0°F, averaged over complete winters (wsi_cold_days from 06c, the Panel C regressor). The 0°F threshold follows the metabolic-stress cutoff in the wildlife literature, not the 32°F road-conditions threshold used elsewhere in the panel. This is the cold component of the Winter Severity Index, without its snow-depth component; see the Winter Severity Index exhibit for the full index.

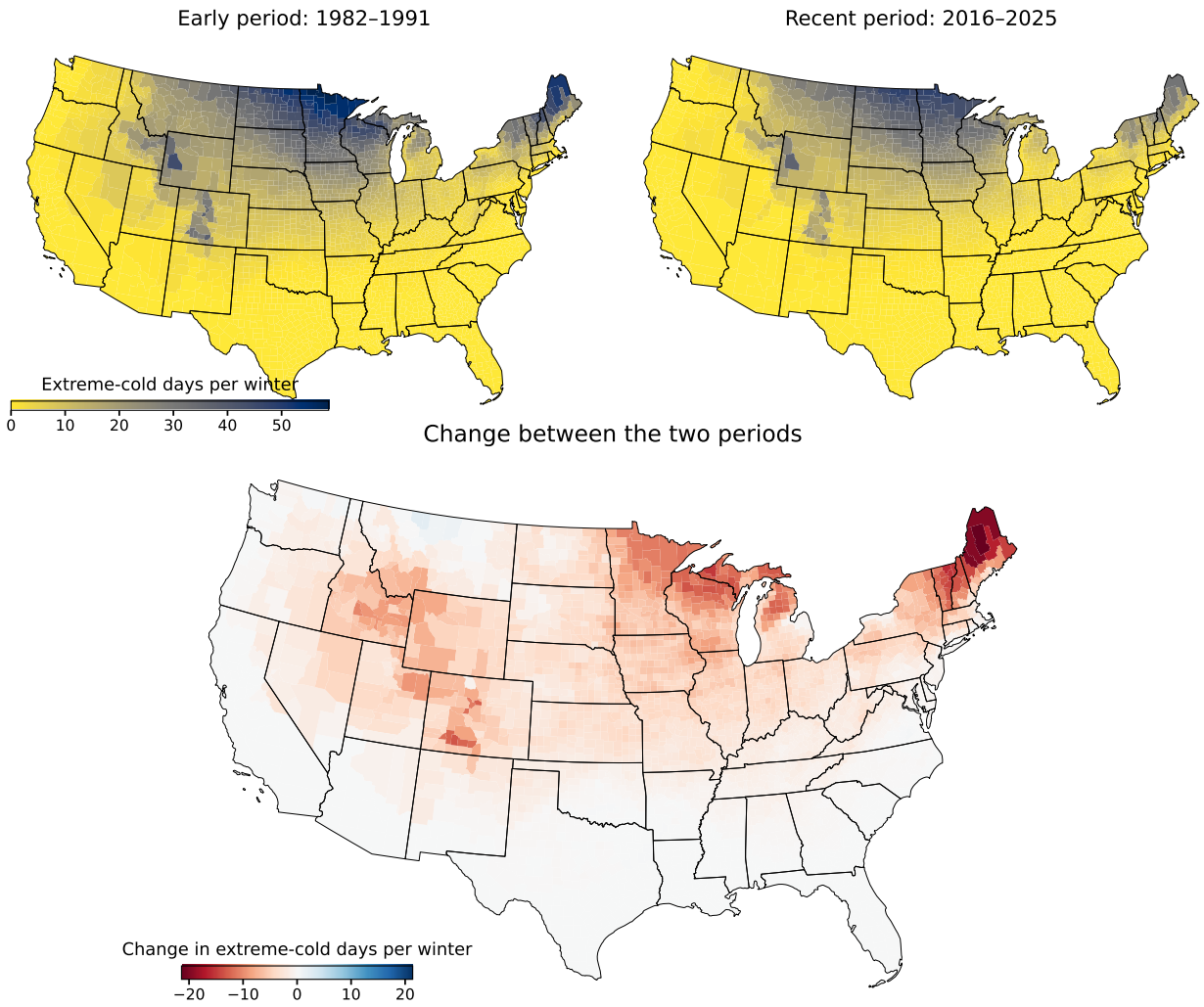


Figure 10: Extreme-cold days (1 December to 30 April) in the early and recent decades, and the change between them.

Notes. Source: PRISM; days from 1 December to 30 April with a daily minimum at or below 0°F, averaged over the complete winters in each period. Top panels share one scale. In the bottom panel RED marks counties that now get FEWER extreme-cold days — the colour scale is deliberately reversed relative to the temperature-change map, because for a count of cold days a decrease is the warming signal. Each county value is a mean over the 10 winters in its period (1982–1991 and 2016–2025), not a comparison of single winters or of sample endpoints.

5 Winter Severity Index

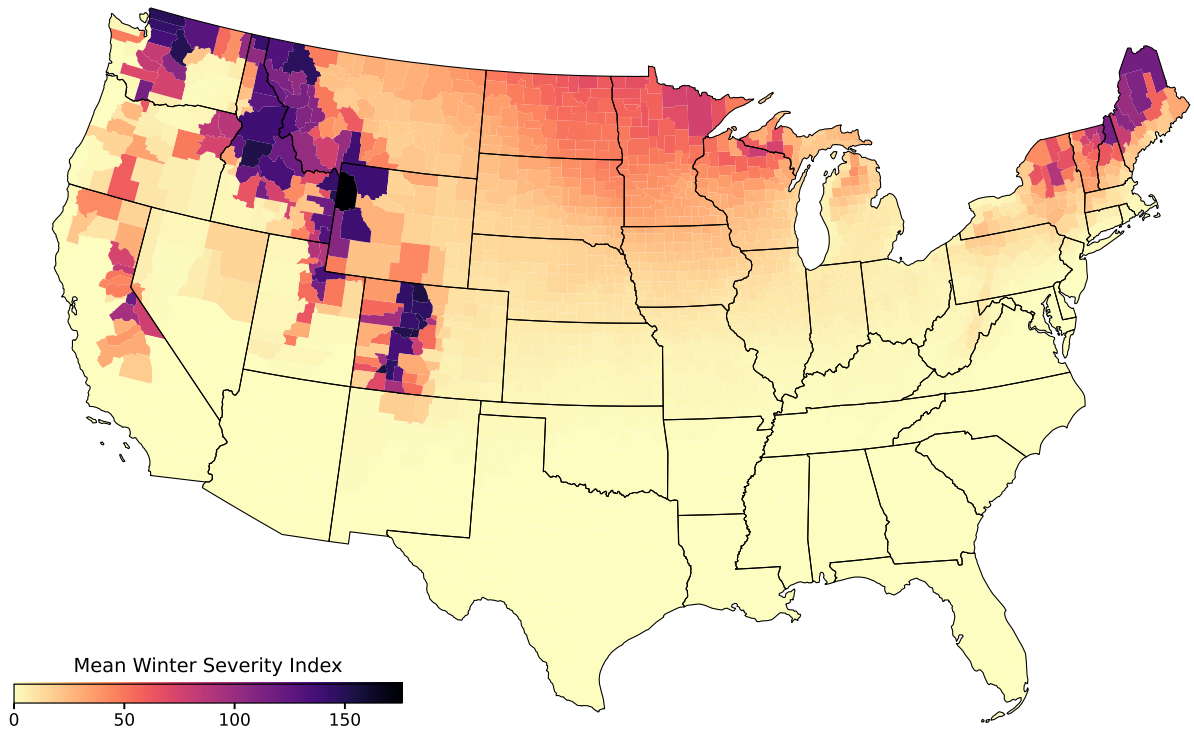


Figure 11: Mean Winter Severity Index (Kohn 1975 / Wisconsin DNR), read from the merged county-year panel.

Notes. Winter Severity Index (Kohn 1975 / Wisconsin DNR): the count of days from 1 December to 30 April with a minimum temperature at or below 0°F, plus the count of days with at least 18 inches of snow on the ground; a day meeting both conditions counts twice. Conventional bands: below 50 mild, 50–80 moderate, 80–100 moderately severe, above 100 very severe. Read from the county-winter-year file built by 06c_build_winter_severity.py, not recomputed here. CAVEAT: snow depth is an ERA5-Land grid-box average averaged again over a county, and that spatial averaging removes the local maxima an 18-inch cutoff is meant to catch, so across most of the eastern and midwestern deer range the index is driven almost entirely by its cold-day component.

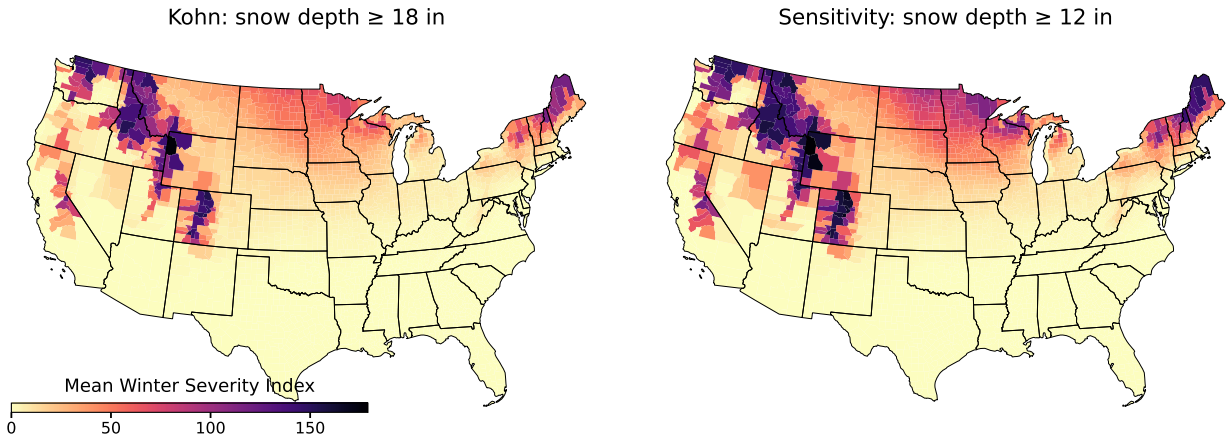


Figure 12: The index under Kohn's 18-inch snow threshold and under a 12-inch alternative, on a shared scale.

Notes. The same index with the snow-hazard component's threshold lowered from Kohn's 18 inches to 12 inches, on a shared colour scale. This is a robustness check, not an alternative definition of the Kohn index: it shows how much of the mapped variation is carried by the cold-day component alone once county-level spatial averaging has flattened the snow signal.

6 Anomalously warm winters

Two thresholds are shown rather than one, and each map is paired with its own time series. The construction matches `warm_winter_1sd` and `warm_winter_2sd` in the merged county-year panel.

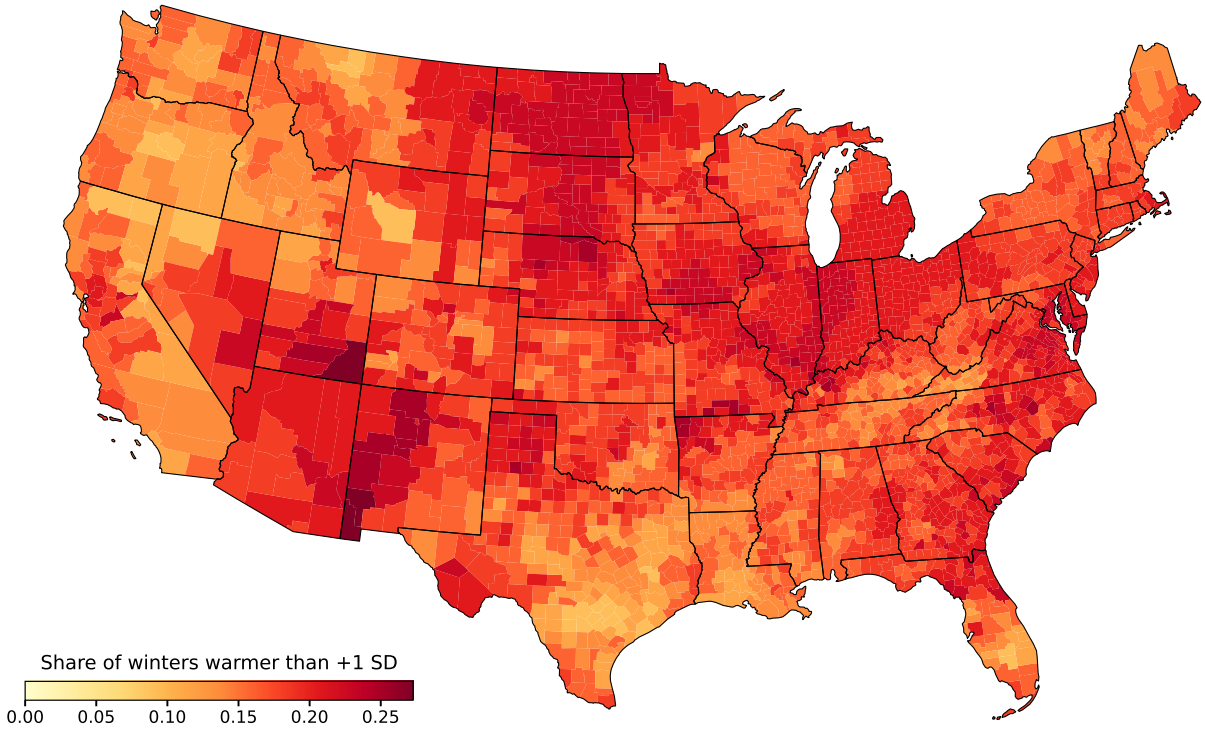


Figure 13: Share of a county's winters more than one standard deviation above its own mean.

Notes. Source: PRISM; complete DJF winters, 1982–2025. A winter is flagged when the county's mean winter temperature exceeds that county's own full-sample mean by more than 1 standard deviations — the identical construction to warm_winter_1sd in the merged county-year panel, so this map describes the regressor used in the collisions model. Colour scales differ between the 1 SD and 2 SD maps; 2 SD winters are far rarer.

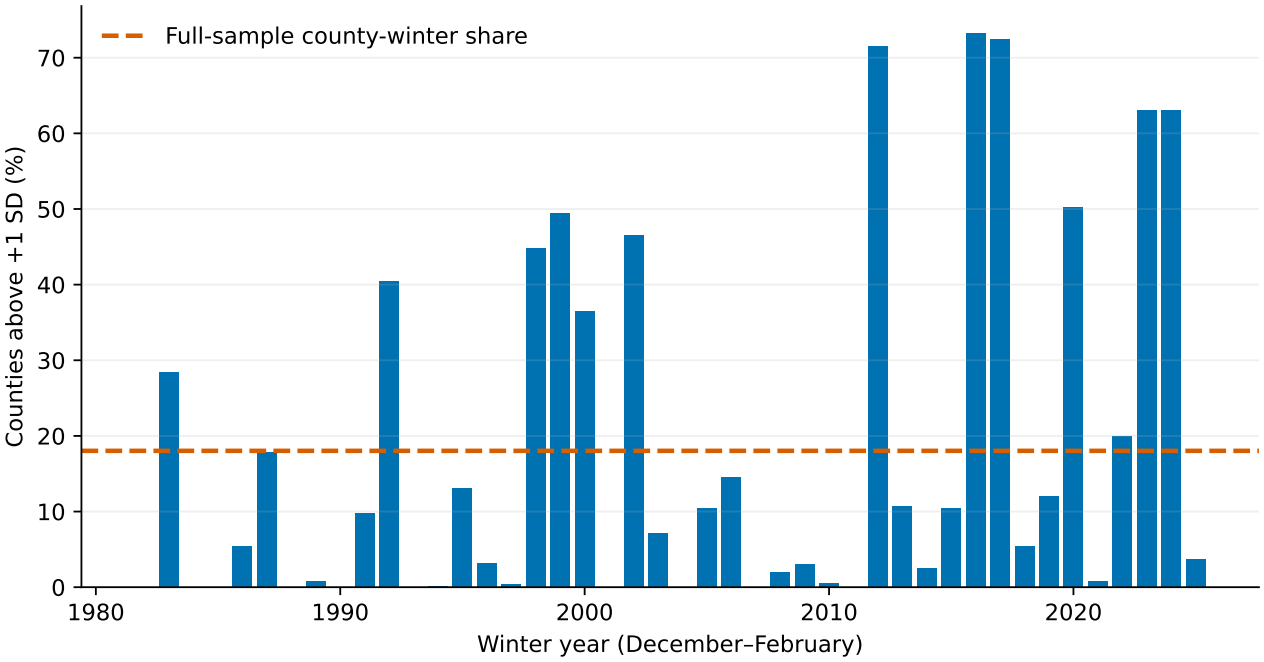


Figure 14: Share of counties flagged at one standard deviation, by winter year.

Notes. Source: PRISM; complete DJF winters. Each bar is the share of counties whose winter mean exceeded that county's own full-sample mean by more than 1 standard deviations. Warm winters arrive as national episodes rather than as a steady drift, and they are spread across the sample rather than concentrated in recent years. Companion to the 1 SD frequency map.

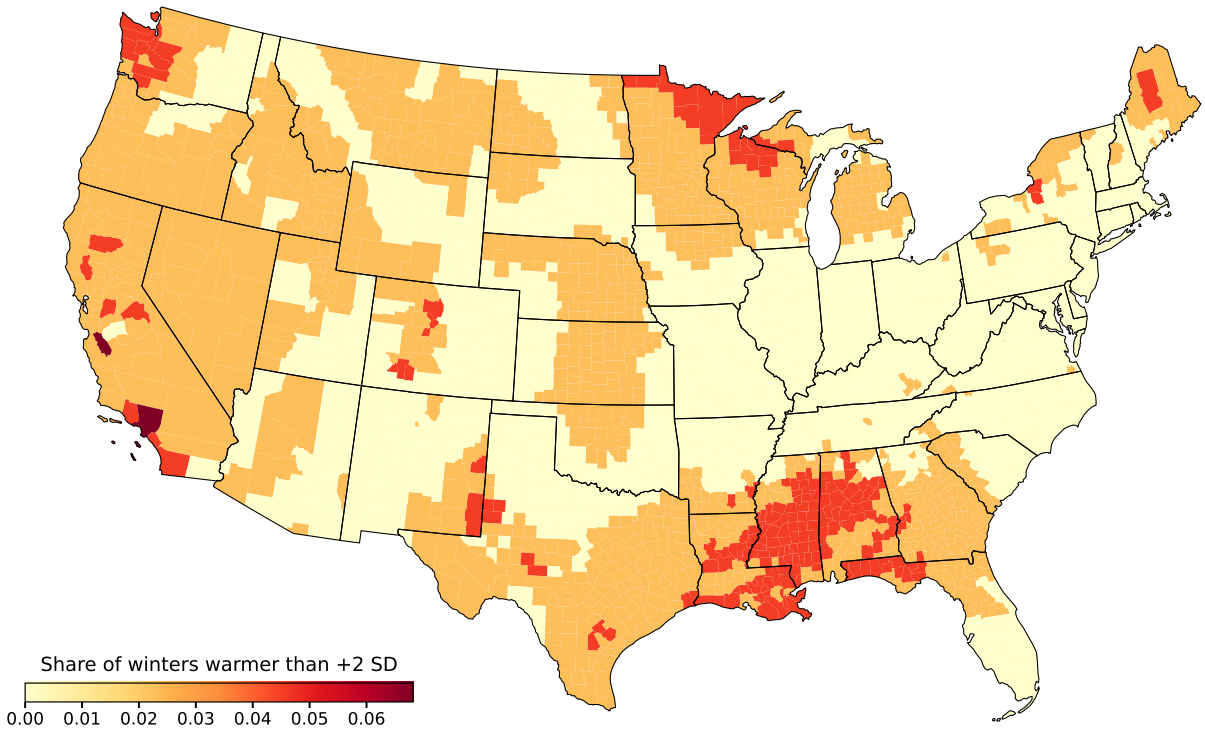


Figure 15: The same map at a two-standard-deviation threshold. Note the independent colour scale.

Notes. Source: PRISM; complete DJF winters, 1982–2025. A winter is flagged when the county's mean winter temperature exceeds that county's own full-sample mean by more than 2 standard deviations — the identical construction to warm_winter_2sd in the merged county-year panel, so this map describes the regressor used in the collisions model. Colour scales differ between the 1 SD and 2 SD maps; 2 SD winters are far rarer.

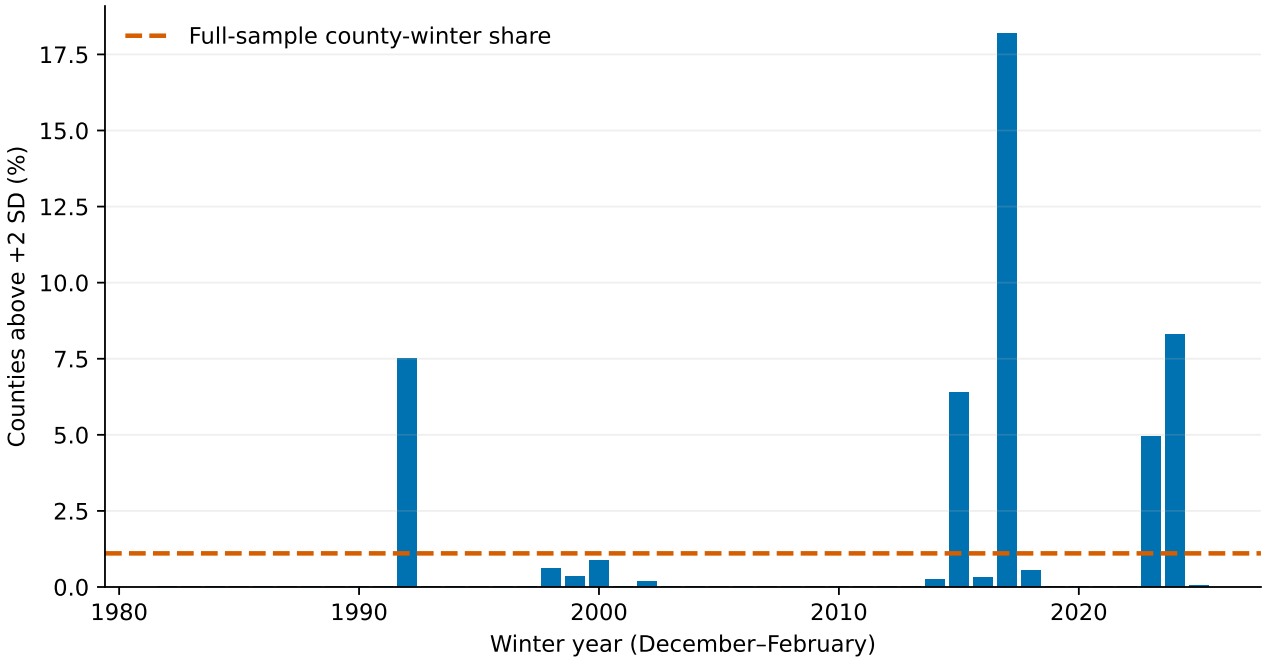


Figure 16: Share of counties flagged at two standard deviations, by winter year.

Notes. Source: PRISM; complete DJF winters. Each bar is the share of counties whose winter mean exceeded that county's own full-sample mean by more than 2 standard deviations. Warm winters arrive as national episodes rather than as a steady drift, and they are spread across the sample rather than concentrated in recent years. Companion to the 2 SD frequency map.

7 Distributional shift

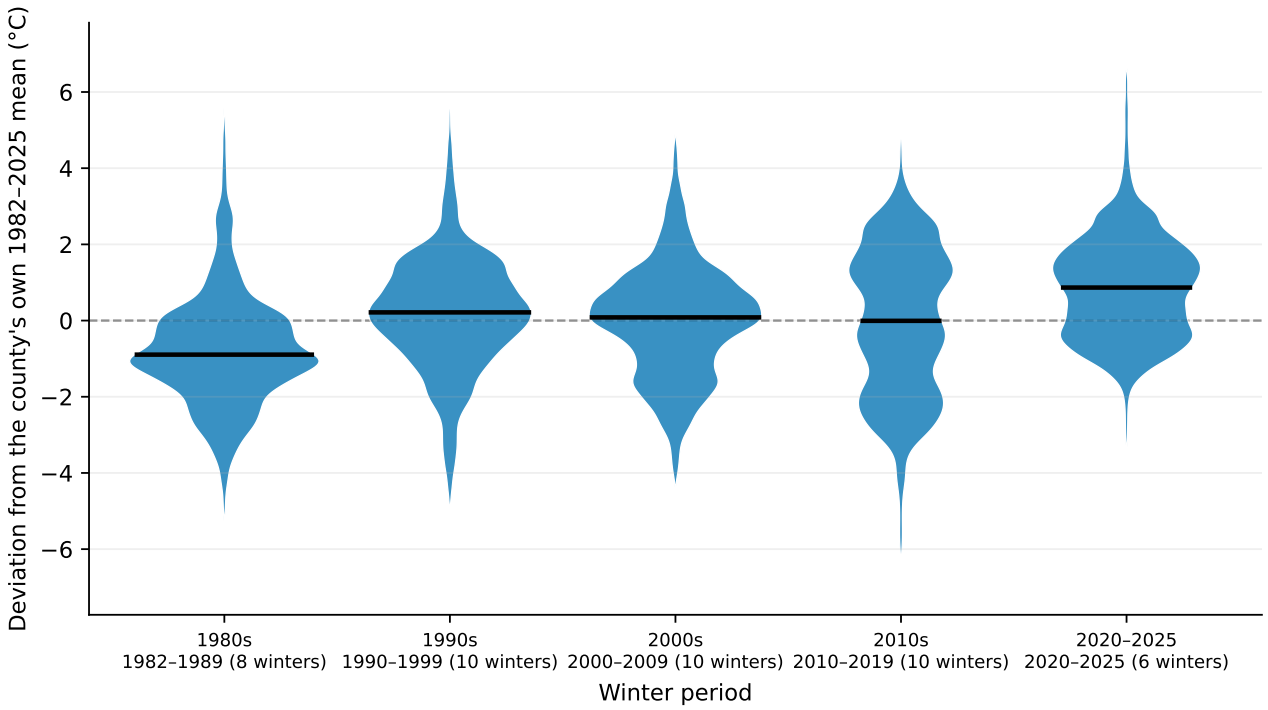


Figure 17: County-winter temperatures by period, after subtracting each county's own full-sample mean.

Notes. Source: PRISM; all complete county-winters, each expressed as a deviation from its own county's full-sample mean. Black bars mark medians; violin width shows density. Removing persistent geography isolates the temporal shift. Periods contain unequal numbers of winters — 1980s: n=24,864; 1990s: n=31,080; 2000s: n=31,080; 2010s: n=31,080; 2020–2025: n=18,648. Estimand: the average county (counties weighted equally), not an area-, population- or road-weighted national average.

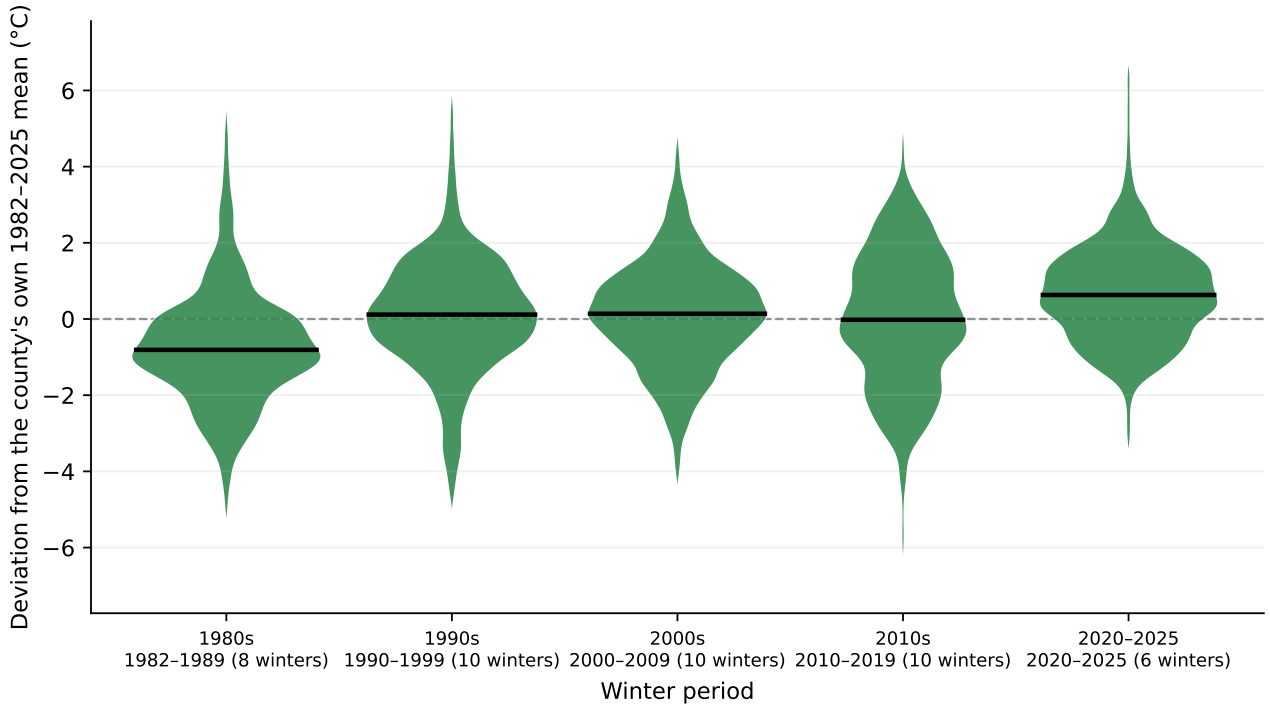


Figure 18: The same distributions with counties weighted by land area.

Notes. Source: PRISM; the same county-winter deviations as the preceding exhibit, with each county weighted by its land area rather than counted once. Densities are weighted Gaussian kernel estimates and medians are weighted medians. Estimand: the average square kilometre of the contiguous United States (counties weighted by land area, population weighting ruled out 8/28); counties in the west are much larger than counties in the east, so the unweighted and area-weighted series answer different questions.

8 Winter warming relative to the other seasons

Whether knowing how much a county warmed in summer tells us how much it warmed in winter. If the seasons had moved together, these would lie on the 45-degree line.

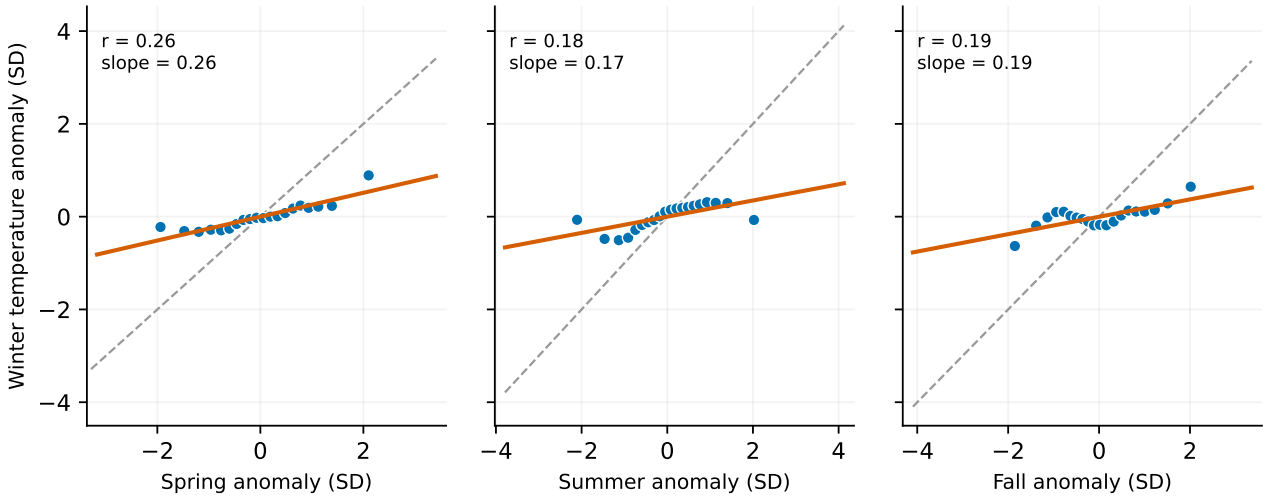


Figure 19: Winter temperature anomaly against the same year's spring, summer and fall anomalies.

Notes. Source: PRISM; one observation per county-year, standardised within county and season against that county-season's own 1982–2025 mean and standard deviation — the same construction as the warm-winter exhibits. Dots are the means of 20 equal-count bins; the orange line is the pooled OLS fit and the grey dashed line is the 45° line. Points on the 45° line would mean a season's anomaly is fully informative about the same year's winter anomaly.

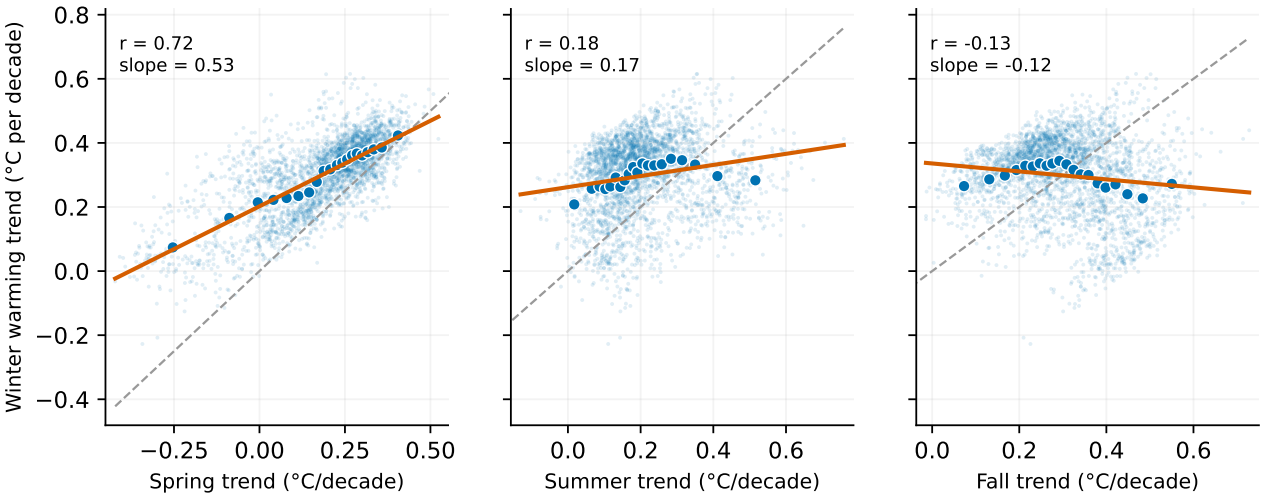


Figure 20: County winter warming trend against that county's spring, summer and fall trends.

Notes. Source: PRISM; one observation per county — the OLS trend in that county's seasonal mean temperature, 1982–2025, in °C per decade. Grey points are counties, coloured dots are the means of 20 equal-count bins, orange is the pooled fit and the grey dashed line is the 45° line. Counties below the 45° line have warmed LESS in winter than in the season on the horizontal axis.

9 Robustness: PRISM versus ERA5

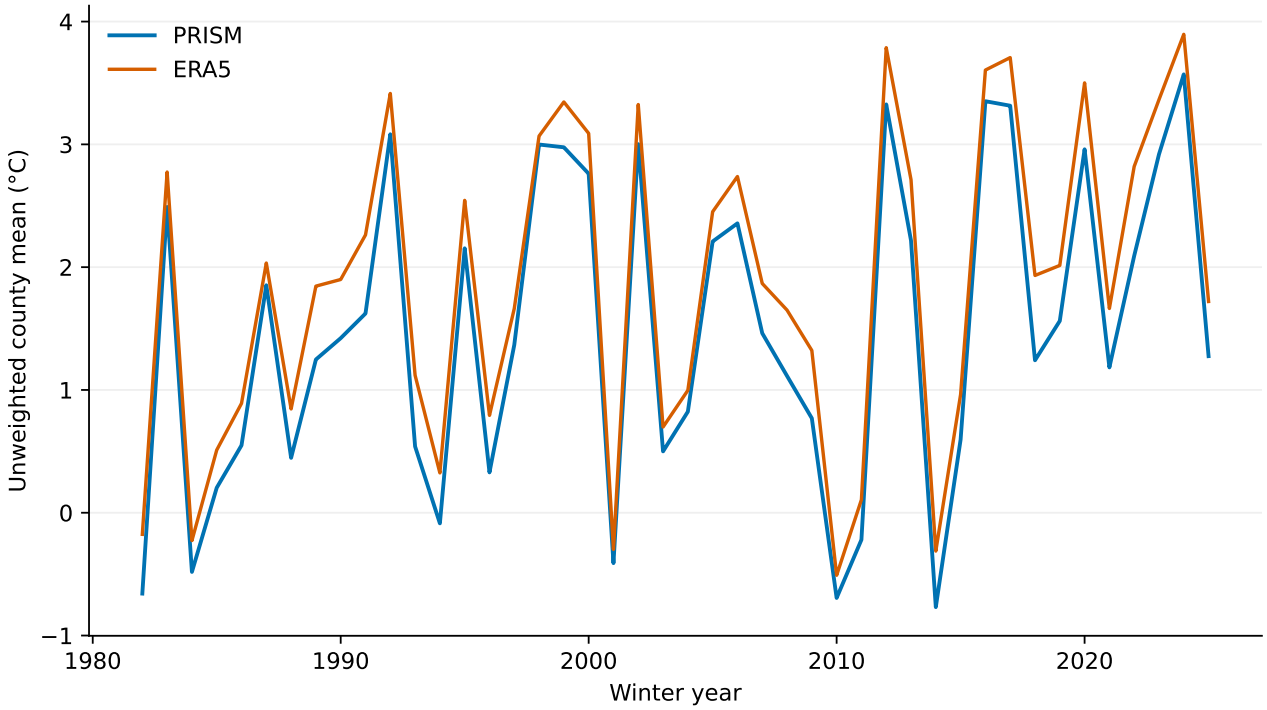


Figure 21: National winter temperature from both products. PRISM is the analysis dataset.

Notes. ERA5 minus PRISM: mean +0.39°C (range +0.07 to +0.73); correlation 0.993. The two products track each other closely and the gap does not widen or close systematically over the sample, which is the property that matters for robustness. PRISM is the analysis dataset; ERA5 is a robustness check only. Estimand: the average county (counties weighted equally), not an area-, population- or road-weighted national average.

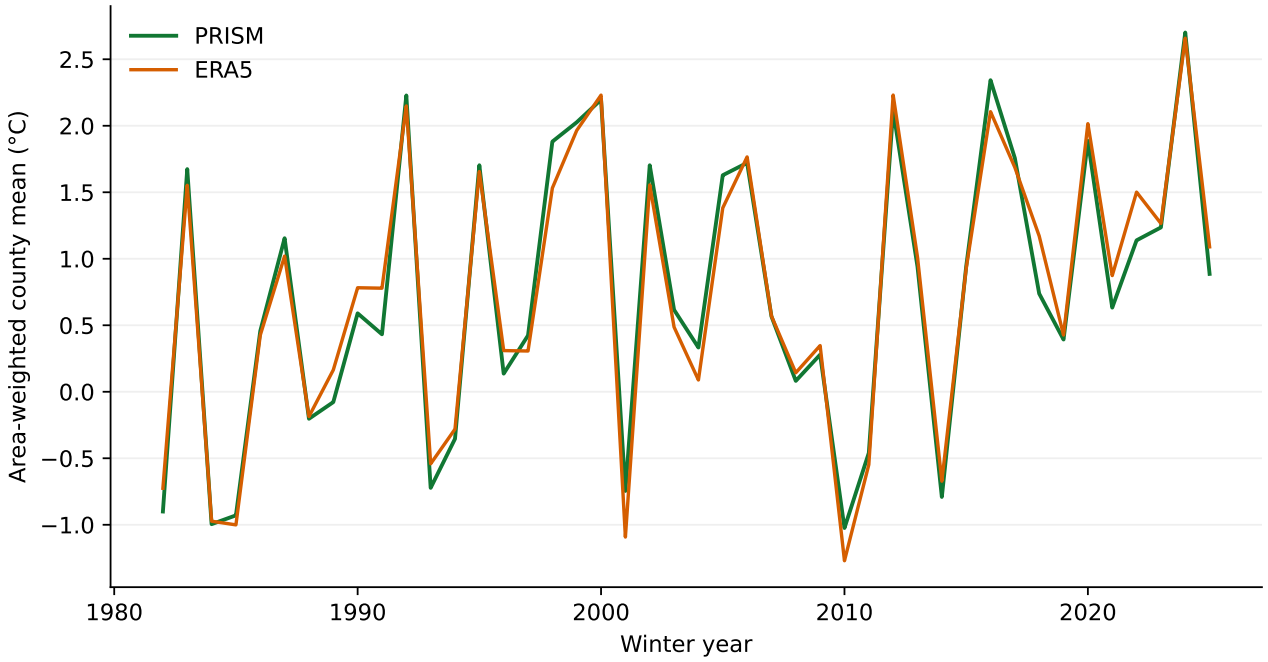
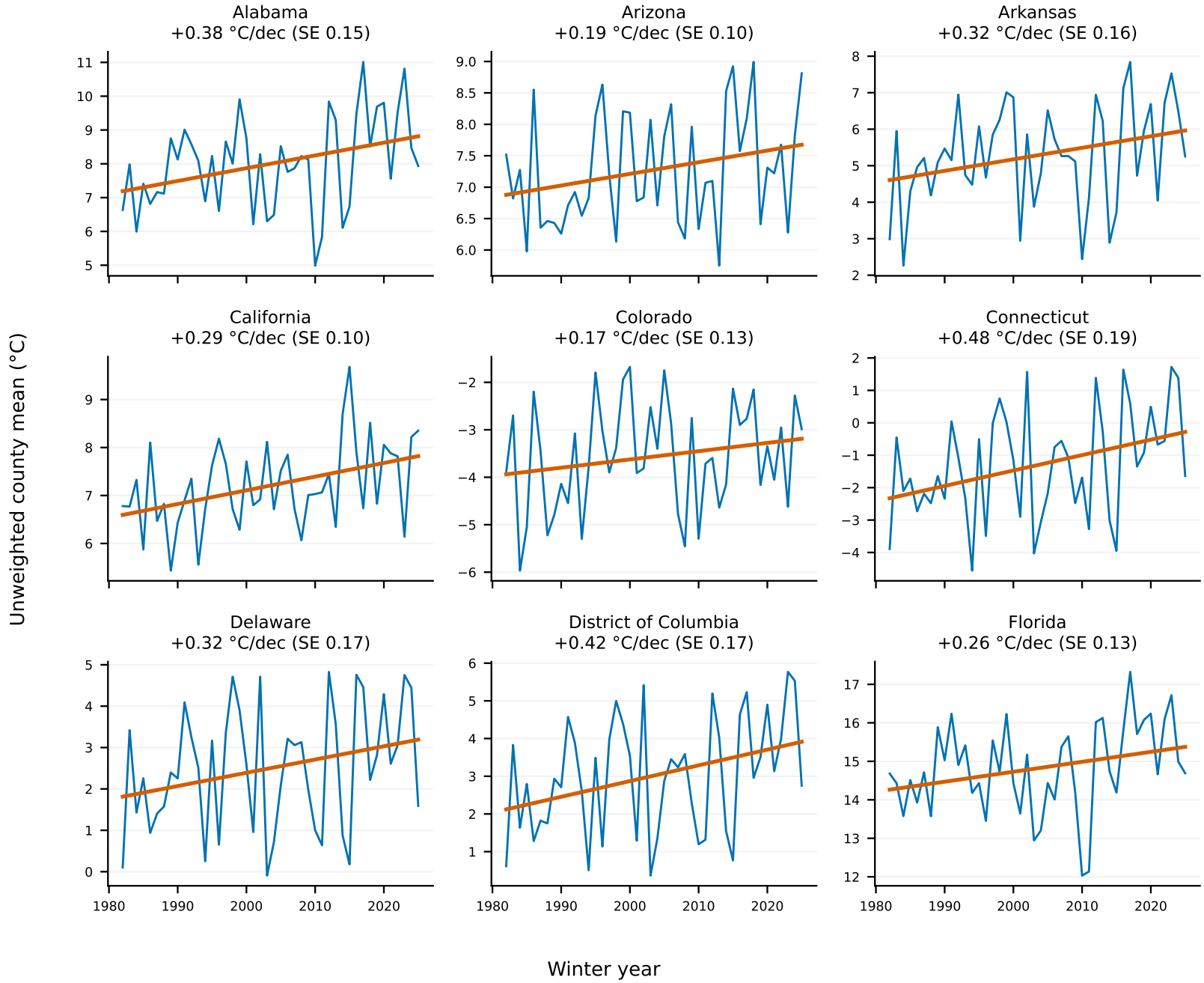


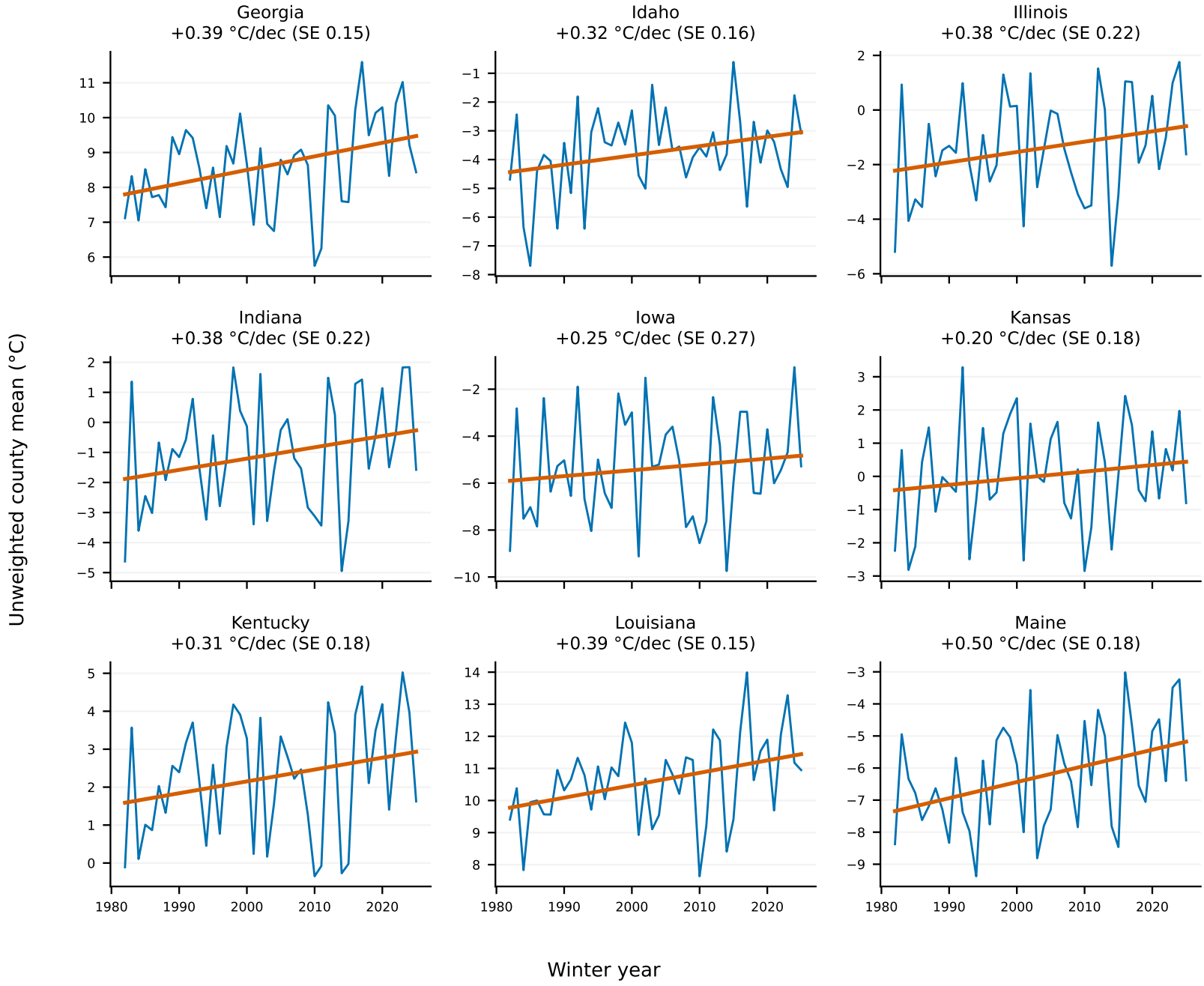
Figure 22: The same comparison with counties weighted by land area.

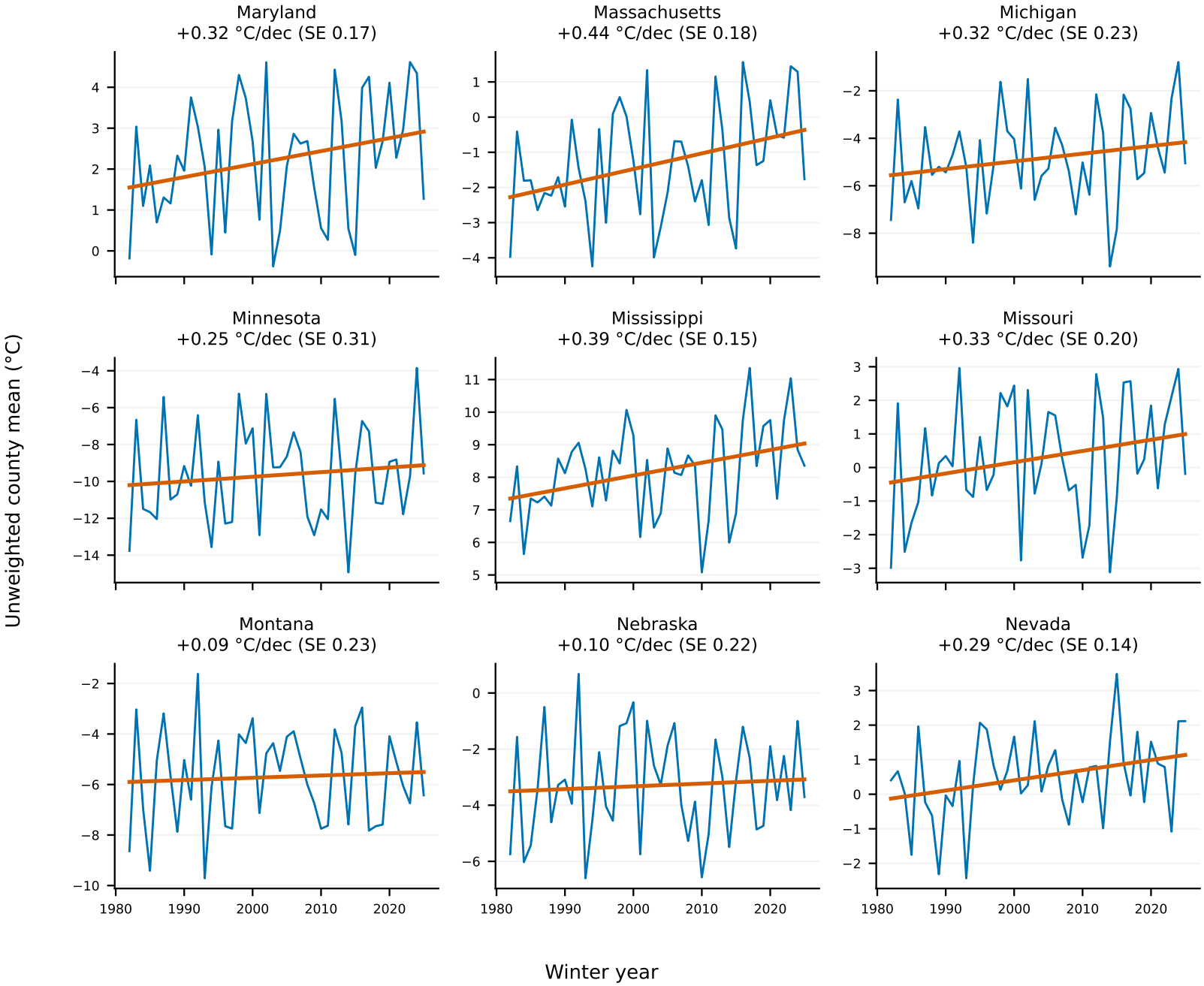
Notes. The same PRISM–ERA5 comparison with counties weighted by land area: mean offset +0.01°C, correlation 0.985. Estimand: the average square kilometre of the contiguous United States (counties weighted by land area, population weighting ruled out 8/28); counties in the west are much larger than counties in the east, so the unweighted and area-weighted series answer different questions.

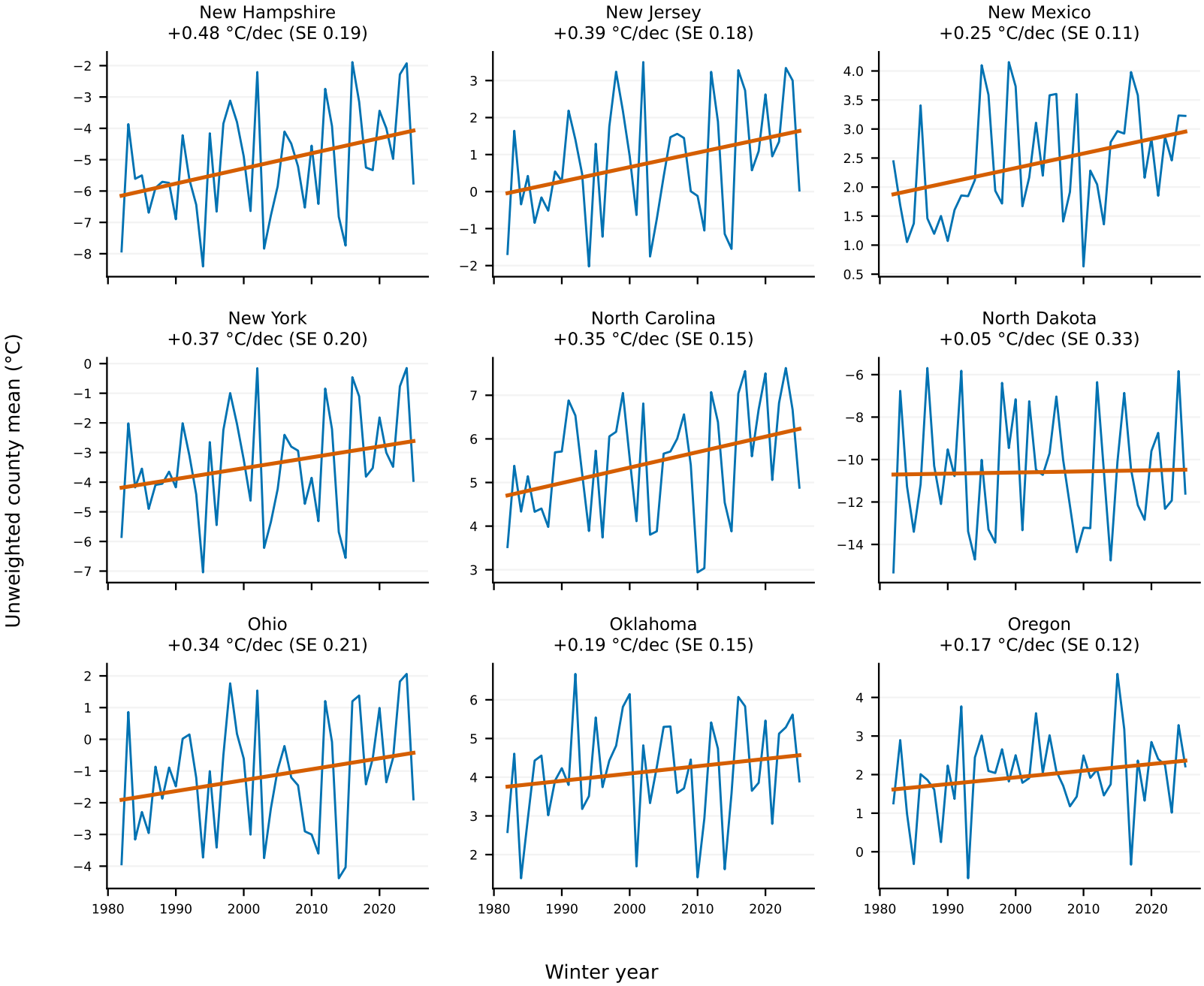
A State-level winter-temperature trends

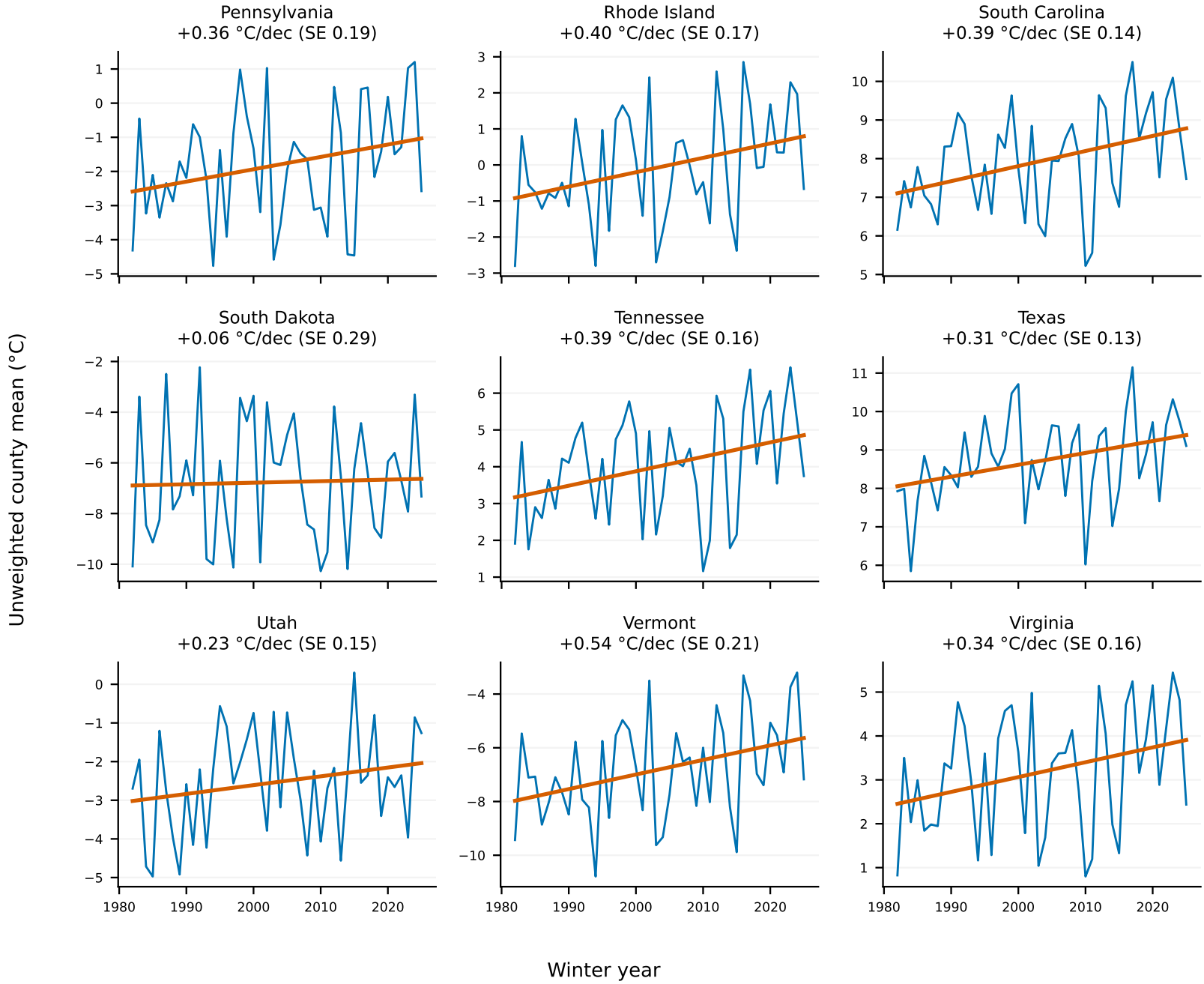
Per-state annual means with state-specific linear trends and the standard error of each slope. Reproduced in full; page count follows the run. Y-axis ranges differ by panel deliberately – a common range across states whose mean winter temperature spans roughly 40°C would flatten every trend to invisibility.

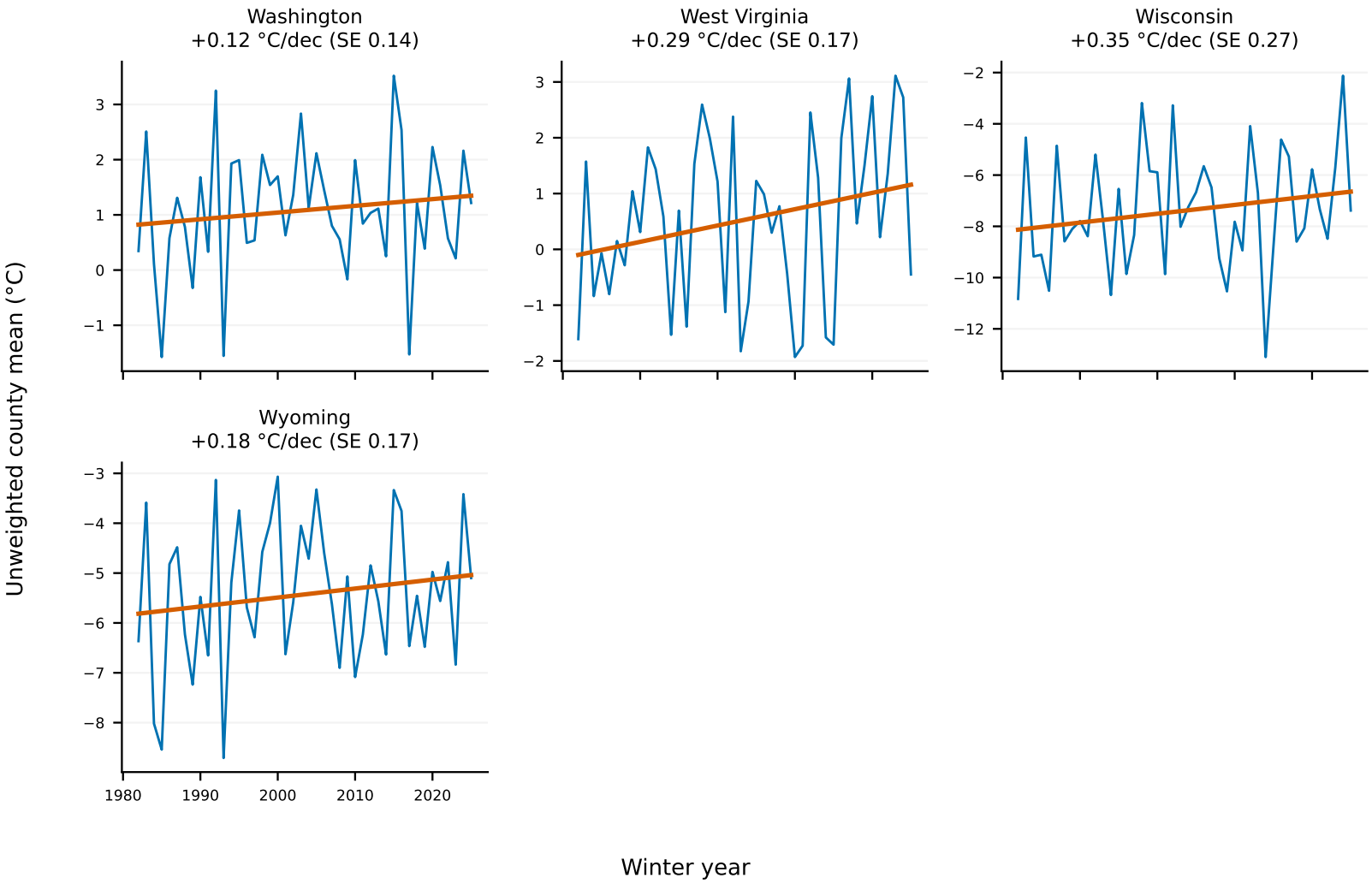

Alabama
+0.38 °C/dec (SE 0.15)
Arizona
+0.19 °C/dec (SE 0.10)
Arkansas
+0.32 °C/dec (SE 0.16)
California
+0.29 °C/dec (SE 0.10)
Colorado
+0.17 °C/dec (SE 0.13)
Connecticut
+0.48 °C/dec (SE 0.19)
Delaware
+0.32 °C/dec (SE 0.17)
District of Columbia
+0.42 °C/dec (SE 0.17)
Florida
+0.26 °C/dec (SE 0.13)
Unweighted county mean (°C)
Winter year


Georgia
+0.39 °C/dec (SE 0.15)
Idaho
+0.32 °C/dec (SE 0.16)
Illinois
+0.38 °C/dec (SE 0.22)
Indiana
+0.38 °C/dec (SE 0.22)
Iowa
+0.25 °C/dec (SE 0.27)
Kansas
+0.20 °C/dec (SE 0.18)
Kentucky
+0.31 °C/dec (SE 0.18)
Louisiana
+0.39 °C/dec (SE 0.15)
Maine
+0.50 °C/dec (SE 0.18)
Unweighted county mean (°C)
Winter year
1980
1990
2000
2010
2020


Maryland
+0.32 °C/dec (SE 0.17)
Massachusetts
+0.44 °C/dec (SE 0.18)
Michigan
+0.32 °C/dec (SE 0.23)
Minnesota
+0.25 °C/dec (SE 0.31)
Mississippi
+0.39 °C/dec (SE 0.15)
Missouri
+0.33 °C/dec (SE 0.20)
Montana
+0.09 °C/dec (SE 0.23)
Nebraska
+0.10 °C/dec (SE 0.22)
Nevada
+0.29 °C/dec (SE 0.14)
Unweighted county mean (°C)
Winter year


New Hampshire
+0.48 °C/dec (SE 0.19)
New Jersey
+0.39 °C/dec (SE 0.18)
New Mexico
+0.25 °C/dec (SE 0.11)
New York
+0.37 °C/dec (SE 0.20)
North Carolina
+0.35 °C/dec (SE 0.15)
North Dakota
+0.05 °C/dec (SE 0.33)
Ohio
+0.34 °C/dec (SE 0.21)
Oklahoma
+0.19 °C/dec (SE 0.15)
Oregon
+0.17 °C/dec (SE 0.12)
Unweighted county mean (°C)
Winter year


Pennsylvania
+0.36 °C/dec (SE 0.19)
Rhode Island
+0.40 °C/dec (SE 0.17)
South Carolina
+0.39 °C/dec (SE 0.14)
South Dakota
+0.06 °C/dec (SE 0.29)
Tennessee
+0.39 °C/dec (SE 0.16)
Texas
+0.31 °C/dec (SE 0.13)
Utah
+0.23 °C/dec (SE 0.15)
Vermont
+0.54 °C/dec (SE 0.21)
Virginia
+0.34 °C/dec (SE 0.16)
Unweighted county mean (°C)
Winter year
1980
1990
2000
2010
2020


Washington
+0.12 °C/dec (SE 0.14)
West Virginia
+0.29 °C/dec (SE 0.17)
Wisconsin
+0.35 °C/dec (SE 0.27)
Wyoming
+0.18 °C/dec (SE 0.17)
Unweighted county mean (°C)
Winter year
1980
1990
2000
2010
2020

B Tier 1 QA artefacts (not reproduced)

The following internal QA exhibits and tables were produced by the same run and are available on request. This includes the pooled-level distribution exhibit, which was demoted from the report set, and the detrended warm-anomaly diagnostic.

- figures/weather/tier1/prism_interannual_variability_choropleth.pdf
- figures/weather/tier1/prism_national_trend_days_below_freezing_32f.pdf
- figures/weather/tier1/prism_national_trend_days_extremely_cold.pdf
- figures/weather/tier1/prism_national_trend_freeze_thaw_days.pdf
- figures/weather/tier1/prism_county_spaghetti_mean_temp_c.pdf
- figures/weather/tier1/prism_state_trends_freeze_thaw_days.pdf
- figures/weather/tier1/prism_state_decade_distributions_mean_temp_c.pdf
- figures/weather/tier1/prism_winter_temperature_distributions_by_decade_tier1.pdf
- figures/weather/tier1/prism_state_trends_wsi_cold_days.pdf
- figures/weather/tier1/prism_national_trend_wsi_cold_days.pdf
- figures/weather/tier1/prism_state_trends_mean_temp_c.pdf
- figures/weather/tier1/prism_state_trends_days_below_freezing_32f.pdf
- figures/weather/tier1/prism_national_trend_mean_temp_c.pdf
- figures/weather/tier1/prism_state_trends_days_extremely_cold.pdf
- tables/weather/tier1/prism_tier1_qa_findings.csv
- tables/weather/tier1/prism_county_winter_warm_anomalies_detrended_diagnostic.csv
- tables/weather/tier1/prism_summary_by_county_review_flags.csv
- tables/weather/tier1/prism_summary_pooled.csv
- tables/weather/tier1/prism_interannual_variability_by_state.csv
- tables/weather/tier1/prism_summary_by_calendar_month.csv
- tables/weather/tier1/prism_tier1_coverage_integrity.csv
- tables/weather/tier1/prism_summary_by_county.csv
- tables/weather/tier1/prism_county_interannual_variability.csv